Strict Moving Horseshoes and Generic Full Complexity in Non-Autonomous Interval Systems

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Abstract

Let $I = [0, 1]$ and let

$$F(I) = C(I, I)^{\mathbb{N}}$$

be the full space of non-autonomous interval systems, equipped with the uniform-in-time metric

$$D(f_{1,\infty}, g_{1,\infty}) = \sup_{n\geq 1} \|f_n - g_n\|_\infty.$$

Balibrea and Rucká asked whether DC1 systems and/or infinite-topological-entropy systems are dense in the full interval non-autonomous space. In [10], the author proved a stronger simultaneous density theorem, including a prescribed dense-open window version. The present paper strengthens that result from density to open density and from infinite entropy to maximal small-scale complexity. For every sequence $(W_n)_{n\geq 1}$ of dense open subsets of I, we construct a D-open dense set $\mathcal{U}(W_\bullet) \subset F(I)$ such that every $g_{1,\infty} \in \mathcal{U}(W_\bullet)$ is DC1, has infinite topological entropy, and satisfies

$$\underline{\mathrm{mdim}}_M(g_{1,\infty}) = \overline{\mathrm{mdim}}_M(g_{1,\infty}) = 1.$$

The value 1 is the largest possible metric mean dimension for interval systems with the Bowen metrics used to define non-autonomous entropy. Moreover, the DC1 scrambled set and the finite separated orbit segments used for entropy and metric mean dimension may be chosen so that their coded orbits visit W_n at time n. The same conclusion holds relatively in the surjective subspace

$$F_s(I) = C_s(I, I)^{\mathbb{N}}.$$

The new ingredient is a strict moving horseshoe: instead of forcing exact relations $g_n(B_{n,i}) = J_{n+1}$, we force robust covering relations with margin. These covering relations survive all sufficiently small uniform-in-time perturbations, which gives the open-dense result.

Keywords. Non-autonomous dynamical systems; distributional chaos; DC1 chaos; topological entropy; metric mean dimension; residual sets; interval maps; moving horseshoes.

1 Introduction

A non-autonomous discrete dynamical system on a compact metric space X is a sequence

$$f_{1,\infty} = (f_1, f_2, f_3, \ldots), \qquad f_n \in C(X, X),$$

with orbit

$$x, \quad f_1(x), \quad f_2(f_1(x)), \quad f_3(f_2(f_1(x))), \ldots.$$

The topological entropy of non-autonomous systems was introduced by Kolyada and Snoha [6]; for interval systems see also Kolyada, Misiurewicz and Snoha [5]. Distributional chaos was introduced by Schweizer and Smítal [8] and has been studied in non-autonomous dynamics in, for example, [4, 9, 11, 1].

Throughout the paper we use the notation of [10]. Thus

$$I = [0,1], \qquad F(I) = C(I,I)^{\mathbb{N}},$$

and, when surjectivity is required,

$$F_s(I) = C_s(I,I)^{\mathbb{N}},$$

where $C_s(I,I)$ denotes the space of continuous surjections $I \to I$. Both spaces are considered with the uniform-in-time metric

$$D(f_{1,\infty}, g_{1,\infty}) = \sup_{n\geq 1} \sup_{x\in I} |f_n(x) - g_n(x)|.$$

This is the same metric denoted ρ_F in parts of the literature, and it is stronger than the product topology because perturbations must be small uniformly in n.

Balibrea and Rucká [1], building on questions from [3, 11], formulated the following problem for the full interval space.

Question 1.1 (Balibrea–Rucká). *Are DC1 and/or infinite-topological-entropy systems dense in $F(I)$?*

The paper [10] answered this affirmatively in simultaneous form. It proved that systems which are both DC1 and of infinite topological entropy are dense in $F(I)$, and also dense in the surjective subspace $F_s(I)$. It also proved a prescribed-window strengthening: for any dense open sets $W_n \subset I$, the local coding intervals can be placed inside W_n, so that the constructed coded orbits satisfy

$$G^{n-1}(x) \in W_n \qquad (n \geq 1).$$

The proof in [10] uses small local horseshoes placed in moving windows. Its basic local relation is

$$g_n(B_{n,i}) = J_{n+1}.$$

The present paper proves that the construction is actually robust. The exact equality above is replaced by a strict covering relation

$$g_n(B_{n,i}) \supset J_{n+1}$$

with a uniform margin. More precisely, if

$$J_{n+1} = [A_{n+1}, B_{n+1}], \qquad B_{n,i} = [r_{n,i}, s_{n,i}],$$

we arrange

$$g_n(r_{n,i}) \leq A_{n+1} - 2\theta, \qquad g_n(s_{n,i}) \geq B_{n+1} + 2\theta$$

for a positive number θ independent of n and i. Therefore, if

$$D(h_{1,\infty}, g_{1,\infty}) < \theta,$$

then

$$h_n(B_{n,i}) \supset J_{n+1}$$

for every n and i. The same symbolic coding, DC1 argument, and entropy argument then work for every system in the whole D-ball around $g_{1,\infty}$.

This gives an open-dense strengthening of [10]. It also strengthens the complexity conclusion. Instead of merely proving infinite entropy, we prove full metric mean dimension. For the non-autonomous Bowen metrics on I, every system has upper metric mean dimension at most 1; our open dense set attains this maximum, and even the lower metric mean dimension is equal to 1.

The result should not be confused with openness questions inside convergent subspaces. Balibrea and Rucká prove that in a pointwise-convergent interval subspace the DC1 systems are neither open nor closed [1]. Our theorem concerns the full non-convergent space $F(I)$ with the uniform-in-time metric. The absence of a convergence constraint is exactly what allows strict moving horseshoes to be implanted at every time.

2 Definitions and main results

For a system $f_{1,\infty} \in F(I)$, write

$$F^0 = \mathrm{id}, \qquad F^n = f_n \circ f_{n-1} \circ \cdots \circ f_1 \quad (n \geq 1).$$

For $N \geq 1$, define the Bowen metric

$$\rho_N(x, y) = \max_{0 \leq j < N} |F^j(x) - F^j(y)|.$$

A set $E \subset I$ is (N, ε)-separated if $\rho_N(x, y) > \varepsilon$ for all distinct $x, y \in E$. Let $s_N(f_{1,\infty}, \varepsilon)$ be the maximal cardinality of an (N, ε)-separated subset of I. Put

$$H_f(\varepsilon) = \limsup_{N\to\infty} \frac{1}{N} \log s_N(f_{1,\infty}, \varepsilon).$$

The topological entropy is

$$h(f_{1,\infty}) = \lim_{\varepsilon \downarrow 0} H_f(\varepsilon) = \sup_{\varepsilon>0} H_f(\varepsilon).$$

The equality with the supremum follows from monotonicity of $H_f(\varepsilon)$ in ε.

We define the lower and upper metric mean dimensions by

$$\underline{\mathrm{mdim}}_M(f_{1,\infty}) = \liminf_{\varepsilon \downarrow 0} \frac{H_f(\varepsilon)}{\log(1/\varepsilon)},$$

and

$$\overline{\mathrm{mdim}}_M(f_{1,\infty}) = \limsup_{\varepsilon \downarrow 0} \frac{H_f(\varepsilon)}{\log(1/\varepsilon)}.$$

These are the separated-set metric mean dimensions associated to the Bowen metrics ρ_N, in the spirit of Lindenstrauss and Weiss [7].

For $x, y \in I$ and $t > 0$, define

$$\psi_{xy}(t) = \liminf_{N\to\infty} \frac{1}{N} \#\{0 \leq j < N : |F^j(x) - F^j(y)| < t\},$$

and

$$\psi^*_{xy}(t) = \limsup_{N\to\infty} \frac{1}{N} \#\{0 \leq j < N : |F^j(x) - F^j(y)| < t\}.$$

The pair (x, y) is DC1-scrambled if

$$\psi^*_{xy}(t) = 1 \quad \text{for every } t > 0$$

and

$$\psi_{xy}(t_0) = 0 \quad \text{for some } t_0 > 0.$$

The system is DC1 if there is an uncountable set $S \subset I$ such that every distinct pair in S is DC1-scrambled.

Our main theorem is the following.

Theorem 2.1 (Open-dense robust window theorem)**.** *Let $(W_n)_{n\geq 1}$ be any sequence of dense open subsets of I. Then there exists a D-open dense set*

$$\mathcal{U}(W_\bullet) \subset F(I)$$

such that every $g_{1,\infty} \in \mathcal{U}(W_\bullet)$ satisfies all of the following conclusions.

1. *$g_{1,\infty}$ is DC1.*
2. *$h(g_{1,\infty}) = \infty$.*
3. *$g_{1,\infty}$ has full metric mean dimension:*

$$\underline{\mathrm{mdim}}_M(g_{1,\infty}) = \overline{\mathrm{mdim}}_M(g_{1,\infty}) = 1.$$

4. *The DC1 scrambled set can be chosen so that, for every scrambled point x and every $n \geq 1$,*

$$G^{n-1}(x) \in W_n,$$

where $G^0 = \mathrm{id}$ and $G^n = g_n \circ \cdots \circ g_1$. The finite separated orbit segments used to prove entropy and metric mean dimension can be chosen with the same window condition.

Moreover,

$$\mathcal{U}(W_\bullet) \cap F_s(I)$$

is relatively open and dense in $F_s(I)$.

Corollary 2.2 (Residual simultaneous chaos and full complexity)**.** *In $F(I)$, the set of systems which are simultaneously DC1, of infinite topological entropy, and of full metric mean dimension contains a dense open set. In particular, it is residual. The same statement holds relatively in $F_s(I)$.*

Remark 2.3 (Strength over [10])**.** *The theorem of [10] says that every D-ball contains a system which is simultaneously DC1 and of infinite entropy, with prescribed dense-open window control. Theorem 2.1 says that every D-ball contains a smaller D-ball all of whose systems have these properties. Thus density is upgraded to open density. In addition, infinite entropy is upgraded to the sharper asymptotic statement $\underline{\mathrm{mdim}}_M = \overline{\mathrm{mdim}}_M = 1$, which is maximal for interval systems.*

3 The block schedule

We use the same block schedule as in [10]. Let

$$\Lambda = \{C\} \cup \{D_r : r \geq 1\} \cup \{E_m : m \geq 2\}.$$

The label C will force long common blocks, the labels D_r will force long disagreement blocks in the r-th binary coordinate, and the labels E_m will force long m-branch entropy blocks.

Choose a sequence

$$\lambda_1, \lambda_2, \lambda_3, \ldots \in \Lambda$$

such that every label in Λ occurs infinitely often. Choose positive integers L_k recursively so fast that, with

$$S_k = L_1 + \cdots + L_{k-1}, \qquad S_1 = 0,$$

one has

$$\frac{S_k}{L_k} \longrightarrow 0.$$

For instance, after $L_1, \ldots, L_{k-1}$ have been chosen, take

$$L_k > k^2(S_k + 1).$$

The k-th time block is

$$T_k = \{S_k + 1, S_k + 2, \ldots, S_k + L_k\}.$$

For $n \geq 1$, let $k(n)$ be the unique index with $n \in T_{k(n)}$. Define the number of branches at time n by

$$M_n = \begin{cases} 2, & \lambda_{k(n)} = C, \\ 2, & \lambda_{k(n)} = D_r \text{ for some } r, \\ m, & \lambda_{k(n)} = E_m. \end{cases}$$

4 Selection inside dense open windows

The perturbation needs many separated points in a window whose images under a given continuous map are close. We state the selection lemma with an explicit separation constant because the dependence on m will later give full metric mean dimension.

Lemma 4.1 (Separated points in a positive-measure set)**.** *Let $A \subset [0,1]$ be Lebesgue measurable and suppose $\lambda(A) > 0$. For every $m \geq 2$, the set A contains m points whose pairwise distances are strictly larger than*

$$\frac{\lambda(A)}{4m}.$$

Proof. Put $\ell = \lambda(A)$ and $r = \ell/(8m)$. Suppose that no m points of A have pairwise distance larger than $2r$. Since every $2r$-separated subset of $[0,1]$ is finite and has uniformly bounded cardinality, there is a $2r$-separated subset $P \subset A$ of maximal cardinality. By assumption, $\mathrm{card}(P) \leq m-1$. By maximality, every point of A lies within distance at most $2r$ of a point of P. Hence A is covered by at most $m-1$ intervals of radius $2r$, each of length at most $4r$. Therefore

$$\lambda(A) \leq (m-1)4r < 4mr = \frac{\ell}{2},$$

which contradicts $\lambda(A) = \ell$. Thus the desired points exist. ☐

Lemma 4.2 (Windowed small-image selection)**.** *Fix $0 < \alpha < 1$ and $m \geq 2$. Put*

$$K_\alpha = \left\lceil \frac{16}{\alpha} \right\rceil, \qquad \delta_m(\alpha) = \frac{\alpha}{64 K_\alpha m}.$$

Let $J \subset I$ be a closed interval of length α, let $W \subset I$ be dense and open, let $f \in C(I, I)$, and let $P \subset I$ be finite. Then there exist points

$$p_1, \dots, p_m \in J \cap W \setminus P$$

such that

$$|p_i - p_j| > 4\delta_m(\alpha) \qquad (i \neq j)$$

and

$$\operatorname{diam}\{f(p_1), \dots, f(p_m)\} < \frac{\alpha}{8}.$$

Moreover, the points may be chosen in the middle closed subinterval of J of length $\alpha/2$.

Proof. Let J^* be the middle closed subinterval of J of length $\alpha/2$. Partition $[0, 1]$ into K_α closed intervals $R_1, \dots, R_{K_\alpha}$ with pairwise disjoint interiors and length at most $\alpha/16$. The measurable sets

$$A_j = J^* \cap f^{-1}(R_j), \qquad 1 \leq j \leq K_\alpha,$$

cover J^* up to overlaps on finitely many endpoints. Therefore, for some j,

$$\lambda(A_j) \geq \frac{|J^*|}{K_\alpha} = \frac{\alpha}{2K_\alpha}.$$

Removing the finite set $P \cup \partial J^*$ does not change the measure. Lemma 4.1 gives points

$$z_1, \dots, z_m \in A_j \setminus (P \cup \partial J^*)$$

whose pairwise distances are larger than

$$\frac{1}{4m} \cdot \frac{\alpha}{2K_\alpha} = \frac{\alpha}{8K_\alpha m} = 8\delta_m(\alpha).$$

All points $f(z_i)$ lie in the same interval R_j, so

$$\operatorname{diam}\{f(z_1), \dots, f(z_m)\} \leq \frac{\alpha}{16}.$$

Since each z_i belongs to the relative interior of J^*, continuity of f gives relatively open intervals $U_i \subset J^*$ around z_i such that

$$U_i \cap P = \varnothing, \qquad \operatorname{dist}(U_i, U_j) > 4\delta_m(\alpha) \quad (i \neq j),$$

and

$$|f(u) - f(z_i)| < \frac{\alpha}{64} \qquad (u \in U_i).$$

Because W is dense and open in I, each $U_i \cap W$ is nonempty. Choose

$$p_i \in U_i \cap W.$$

Then $p_i \in J \cap W \setminus P$, the points are pairwise more than $4\delta_m(\alpha)$ apart, and

$$\operatorname{diam}\{f(p_1), \dots, f(p_m)\} \leq \frac{\alpha}{16} + 2 \cdot \frac{\alpha}{64} = \frac{3\alpha}{32} < \frac{\alpha}{8}.$$

□

5 Strict windowed horseshoe schemes

We now define the robust structure which replaces the exact coding relation of [10].

Definition 5.1 (Strict $W_\bullet$-windowed horseshoe scheme)**.** *Let $(W_n)_{n\geq 1}$ be dense open subsets of I. A system $g_{1,\infty} \in F(I)$ admits a strict $W_\bullet$-windowed horseshoe scheme if there exist numbers*

$$0 < \alpha < 1, \qquad \theta > 0,$$

closed intervals

$$J_n = [A_n, B_n] \subset [2\theta, 1 - 2\theta], \qquad |J_n| = \alpha,$$

and, for each $n \geq 1$, closed nondegenerate branch intervals

$$B_{n,1}, \ldots, B_{n,M_n} \subset J_n \cap W_n, \qquad B_{n,i} = [r_{n,i}, s_{n,i}],$$

such that:

1. $\operatorname{diam}(B_{n,i}) < 1/n$ *for every n and i;*
2. $\operatorname{dist}(B_{n,i}, B_{n,j}) > \delta_{M_n}(\alpha)$ *whenever $i \neq j$;*
3. *for every n and i,*
$$g_n(r_{n,i}) \leq A_{n+1} - 2\theta, \qquad g_n(s_{n,i}) \geq B_{n+1} + 2\theta.$$

Let $\mathcal{U}(W_\bullet)$ denote the set of all systems in $F(I)$ which admit such a scheme.

Lemma 5.2 (Openness of strict schemes)**.** *For every sequence $(W_n)_{n\geq 1}$ of dense open subsets of I, the set $\mathcal{U}(W_\bullet)$ is open in $F(I)$.*

Proof. Let $g_{1,\infty} \in \mathcal{U}(W_\bullet)$, and fix a strict scheme for it with margin θ. Suppose

$$D(h_{1,\infty}, g_{1,\infty}) < \theta.$$

For every n and i,

$$h_n(r_{n,i}) < A_{n+1} - \theta = A_{n+1} - 2(\theta/2),$$

and

$$h_n(s_{n,i}) > B_{n+1} + \theta = B_{n+1} + 2(\theta/2).$$

Also

$$J_n \subset [2\theta, 1 - 2\theta] \subset [\theta, 1 - \theta].$$

Thus the same intervals J_n and $B_{n,i}$ form a strict $W_\bullet$-windowed horseshoe scheme for $h_{1,\infty}$, with margin $\theta/2$. Hence the D-ball of radius θ around $g_{1,\infty}$ is contained in $\mathcal{U}(W_\bullet)$. □

Lemma 5.3 (Robust covering)**.** *Let $g_{1,\infty}$ admit a strict scheme with margin θ. If*

$$D(h_{1,\infty}, g_{1,\infty}) < \theta,$$

then

$$h_n(B_{n,i}) \supset J_{n+1}$$

for every n and i. In particular, taking $h_{1,\infty} = g_{1,\infty}$ gives

$$g_n(B_{n,i}) \supset J_{n+1}.$$

Proof. Write
$$J_{n+1} = [A_{n+1}, B_{n+1}], \qquad B_{n,i} = [r_{n,i}, s_{n,i}].$$
If $D(h, g) < \theta$, then
$$h_n(r_{n,i}) < A_{n+1}, \qquad h_n(s_{n,i}) > B_{n+1}.$$
Since h_n is continuous on $[r_{n,i}, s_{n,i}]$, the intermediate value theorem gives
$$h_n([r_{n,i}, s_{n,i}]) \supset [A_{n+1}, B_{n+1}].$$
For $h = g$ the same conclusion follows directly from the strict inequalities with margin 2θ. $\square$

6 Dense implantation of strict schemes

The next lemma is the perturbative core of the paper. It is the strict analogue of the local perturbation scheme in [10].

Lemma 6.1 (Dense strict implantation)**.** *Let $(W_n)_{n\geq 1}$ be dense open subsets of I. For every $f_{1,\infty} \in F(I)$ and every $\eta > 0$, there exists*
$$g_{1,\infty} \in \mathcal{U}(W_\bullet)$$
such that
$$D(f_{1,\infty}, g_{1,\infty}) < \eta.$$
If $f_{1,\infty} \in F_s(I)$, then $g_{1,\infty}$ may be chosen in $F_s(I)$.

Proof. Fix $f_{1,\infty} \in F(I)$ and $\eta > 0$. Put
$$\alpha = \frac{\min\{\eta, 1\}}{100}, \qquad \theta = \frac{\alpha}{20}.$$
Then $0 < \alpha < 1$ and
$$\alpha + 4\theta < 1,$$
so intervals of length α fit inside $[2\theta, 1 - 2\theta]$.

We recursively construct closed intervals
$$J_n = [A_n, B_n] \subset [2\theta, 1 - 2\theta], \qquad |J_n| = \alpha,$$
auxiliary intervals $D_{n,i}$, branch intervals $B_{n,i}$, and maps g_n.

Start with
$$J_1 = [2\theta, 2\theta + \alpha].$$
Assume J_n has been chosen. Put $m = M_n$. If we are proving the surjective version and f_n is surjective, choose points $a_n, b_n \in I$ such that
$$f_n(a_n) = 0, \qquad f_n(b_n) = 1,$$
and put
$$P_n = \{a_n, b_n\}.$$
If surjectivity preservation is not required, put $P_n = \varnothing$.

Apply Lemma 4.2 to J_n, W_n, f_n, P_n, and m. We obtain points

$$p_{n,1}, \dots, p_{n,m} \in J_n \cap W_n \setminus P_n$$

such that

$$|p_{n,i} - p_{n,j}| > 4\delta_m(\alpha) \quad (i \neq j)$$

and

$$\operatorname{diam}\{f_n(p_{n,1}), \dots, f_n(p_{n,m})\} < \frac{\alpha}{8}.$$

Because f_n is continuous, W_n is open, the points $p_{n,i}$ are separated, and none of them belongs to P_n, we may choose pairwise disjoint closed intervals

$$D_{n,1}, \dots, D_{n,m} \subset J_n \cap W_n$$

with

$$p_{n,i} \in \operatorname{int}(D_{n,i}), \qquad D_{n,i} \cap P_n = \varnothing,$$
$$\operatorname{dist}(D_{n,i}, D_{n,j}) > 2\delta_m(\alpha) \quad (i \neq j),$$

and

$$\operatorname{diam} f_n\left(D_{n,1} \cup \cdots \cup D_{n,m}\right) < \frac{\alpha}{4}.$$

Now choose smaller closed nondegenerate intervals

$$B_{n,i} = [r_{n,i}, s_{n,i}] \subset \operatorname{int}(D_{n,i})$$

so that

$$\operatorname{diam}(B_{n,i}) < \frac{1}{n}, \qquad \operatorname{dist}(B_{n,i}, B_{n,j}) > \delta_m(\alpha) \quad (i \neq j).$$

Let

$$c_{n+1} = f_n(p_{n,1}).$$

Choose $J_{n+1} = [A_{n+1}, B_{n+1}]$ by setting

$$A_{n+1} = \min\left\{\max\left\{c_{n+1} - \frac{\alpha}{2}, 2\theta\right\}, 1 - 2\theta - \alpha\right\},$$

and

$$B_{n+1} = A_{n+1} + \alpha.$$

Then

$$J_{n+1} \subset [2\theta, 1 - 2\theta], \qquad |J_{n+1}| = \alpha.$$

Moreover, every point of the enlarged interval

$$[A_{n+1} - 2\theta, B_{n+1} + 2\theta]$$

is within distance at most $\alpha + 4\theta$ of c_{n+1}. Indeed, if no truncation occurs, this is immediate. If lower truncation occurs, then $c_{n+1} \leq 2\theta + \alpha/2$, and the right endpoint $B_{n+1} + 2\theta = \alpha + 4\theta$ is at distance at most $\alpha + 4\theta$ from c_{n+1}, while the left endpoint is 0. The upper truncation case is symmetric.

We now define g_n. Outside $D_{n,1} \cup \cdots \cup D_{n,m}$, set

$$g_n(x) = f_n(x).$$

Fix i. Write
$$D_{n,i} = [u,v], \qquad B_{n,i} = [r,s], \qquad J_{n+1} = [A,B].$$
On $D_{n,i}$, define g_n to be the piecewise linear map determined by
$$g_n(u) = f_n(u), \qquad g_n(r) = A - 2\theta,$$
$$g_n(s) = B + 2\theta, \qquad g_n(v) = f_n(v),$$
and by linearity on $[u,r]$, $[r,s]$, and $[s,v]$. Since $J_{n+1} \subset [2\theta, 1-2\theta]$, the values $A-2\theta$ and $B+2\theta$ lie in $[0,1]$. The pieces glue continuously because $g_n = f_n$ at the endpoints of each $D_{n,i}$. Thus $g_n \in C(I,I)$.

We estimate the perturbation. If $x \notin \bigcup_i D_{n,i}$, then $g_n(x) = f_n(x)$. If $x \in D_{n,i}$, then by construction
$$|f_n(x) - c_{n+1}| < \frac{\alpha}{4}.$$
All values used to define g_n on $D_{n,i}$ are within distance at most $\alpha + 4\theta$ of c_{n+1}: this is true for $A-2\theta$ and $B+2\theta$ by the choice of J_{n+1}, and it is true for $f_n(u)$ and $f_n(v)$ by the preceding display. Since $g_n(x)$ is a convex combination of the relevant endpoint values on each linear piece,
$$|g_n(x) - c_{n+1}| \le \alpha + 4\theta.$$
Therefore
$$|g_n(x) - f_n(x)| < \alpha + 4\theta + \frac{\alpha}{4} = \frac{5\alpha}{4} + 4\theta = \frac{29\alpha}{20} < 2\alpha.$$
Taking the supremum over x and n gives
$$D(f_{1,\infty}, g_{1,\infty}) < 2\alpha < \eta.$$

Finally,
$$g_n(r_{n,i}) = A_{n+1} - 2\theta, \qquad g_n(s_{n,i}) = B_{n+1} + 2\theta.$$
Thus $g_{1,\infty}$ admits a strict $W_\bullet$-windowed horseshoe scheme, so $g_{1,\infty} \in \mathcal{U}(W_\bullet)$.

It remains to check surjectivity. Suppose $f_{1,\infty} \in F_s(I)$. Then each f_n is surjective, and the modification intervals were chosen disjoint from $P_n = \{a_n, b_n\}$. Hence
$$g_n(a_n) = f_n(a_n) = 0, \qquad g_n(b_n) = f_n(b_n) = 1.$$
The continuous map $g_n : I \to I$ attains both 0 and 1, so by the intermediate value theorem $g_n(I) = I$. Thus every g_n is surjective, and $g_{1,\infty} \in F_s(I)$. □

7 Symbolic coding from covering relations

Let $g_{1,\infty} \in \mathcal{U}(W_\bullet)$, and fix a strict scheme. Write
$$G^0 = \mathrm{id}, \qquad G^n = g_n \circ g_{n-1} \circ \cdots \circ g_1 \quad (n \ge 1).$$
By Lemma 5.3,
$$g_n(B_{n,i}) \supset J_{n+1} \qquad (1 \le i \le M_n).$$
Only this covering relation is used in the rest of the proof.

Lemma 7.1 (Infinite itinerary coding)**.** *For every sequence*

$$\omega = (\omega_1, \omega_2, \ldots), \qquad \omega_n \in \{1, \ldots, M_n\},$$

there exists $x_\omega \in J_1$ such that

$$G^{n-1}(x_\omega) \in B_{n,\omega_n} \subset W_n \qquad (n \geq 1).$$

Moreover, if two itineraries differ at some time n, then no point realizes both itineraries.

Proof. First fix $N \geq 1$. We prove that there exists $x \in J_1$ such that

$$G^{n-1}(x) \in B_{n,\omega_n} \qquad (1 \leq n \leq N).$$

Choose any $y_N \in B_{N,\omega_N}$. Since $y_N \in J_N$ and

$$g_{N-1}(B_{N-1,\omega_{N-1}}) \supset J_N$$

when $N \geq 2$, there exists $y_{N-1} \in B_{N-1,\omega_{N-1}}$ such that $g_{N-1}(y_{N-1}) = y_N$. Repeating this backward step gives $y_1 \in B_{1,\omega_1} \subset J_1$ with the required finite itinerary. For $N = 1$, this simply means choosing $y_1 \in B_{1,\omega_1}$.

Thus the compact sets

$$K_N(\omega) = \{x \in J_1 : G^{n-1}(x) \in B_{n,\omega_n} \text{ for } 1 \leq n \leq N\}$$

are nonempty, closed, and nested. Their intersection is nonempty by compactness of J_1. Every point in the intersection realizes the infinite itinerary.

If two itineraries differ at time n, then a point realizing both would have $G^{n-1}(x)$ in two distinct branch intervals $B_{n,i}$ and $B_{n,j}$. These intervals are disjoint because their distance is positive. This is impossible. □

Corollary 7.2 (Finite itinerary coding)**.** *For every $N \geq 1$ and every finite admissible word*

$$(\omega_1, \ldots, \omega_N), \qquad \omega_n \in \{1, \ldots, M_n\},$$

there exists $x \in J_1$ such that

$$G^{n-1}(x) \in B_{n,\omega_n} \qquad (1 \leq n \leq N).$$

Proof. This is the finite-itinerary assertion proved in the first paragraph of Lemma 7.1. □

8 DC1 chaos

We now define an uncountable family of itineraries exactly as in [10]. Let

$$\beta = (\beta_1, \beta_2, \ldots) \in \{0, 1\}^{\mathbb{N}}.$$

Define an itinerary ω^β block by block.

If $n \in T_k$ and $\lambda_k = C$, set

$$\omega_n^\beta = 1.$$

Thus all itineraries agree on every C-block.

If $n \in T_k$ and $\lambda_k = D_r$, set

$$\omega_n^\beta = 1 + \beta_r.$$

Thus on a D_r-block the itinerary records the r-th binary digit of β, constantly across the whole block.

If $n \in T_k$ and $\lambda_k = E_m$, set

$$\omega_n^\beta = 1.$$

Entropy blocks are ignored by the DC1 coding.

For each $\beta \in \{0,1\}^{\mathbb{N}}$, choose one point

$$x_\beta = x_{\omega^\beta}$$

supplied by Lemma 7.1, and put

$$S = \{x_\beta : \beta \in \{0,1\}^{\mathbb{N}}\}.$$

Proposition 8.1 (Windowed DC1 scrambled set). *The set S is uncountable, every distinct pair of points in S is DC1-scrambled, and*

$$G^{n-1}(x) \in W_n \qquad (x \in S,\ n \geq 1).$$

In particular, $g_{1,\infty}$ is DC1.

Proof. The window condition follows directly from Lemma 7.1. We first show that S is uncountable. If $\beta \neq \gamma$, choose $r \geq 1$ such that $\beta_r \neq \gamma_r$. Since D_r occurs infinitely often, there is a block T_k with $\lambda_k = D_r$. On that block, ω^β and ω^γ take different values. If $x_\beta = x_\gamma$, then one point would realize two itineraries which differ at some time, contradicting Lemma 7.1. Thus $\beta \mapsto x_\beta$ is injective, so S is uncountable.

Now fix distinct β, γ. We prove the two DC1 conditions. Let $t > 0$. Since

$$\operatorname{diam}(B_{n,i}) < \frac{1}{n},$$

there exists N_t such that $\operatorname{diam}(B_{n,i}) < t$ for every $n \geq N_t$ and every admissible i. Consider a block T_k with $\lambda_k = C$ and $S_k + 1 \geq N_t$. There are infinitely many such blocks. For every $n \in T_k$, both itineraries have value 1, so

$$G^{n-1}(x_\beta),\ G^{n-1}(x_\gamma) \in B_{n,1}.$$

Therefore

$$|G^{n-1}(x_\beta) - G^{n-1}(x_\gamma)| < t \qquad (n \in T_k).$$

Equivalently, the two orbits are t-close for the L_k iterates

$$j = S_k, S_k + 1, \ldots, S_k + L_k - 1.$$

Hence

$$\frac{1}{S_k + L_k} \#\{0 \leq j < S_k + L_k : |G^j(x_\beta) - G^j(x_\gamma)| < t\} \geq \frac{L_k}{S_k + L_k}.$$

Along the infinitely many C-blocks, $S_k/L_k \to 0$, and the right-hand side tends to 1. Thus

$$\psi^*_{x_\beta x_\gamma}(t) = 1.$$

Since $t > 0$ was arbitrary,

$$\psi^*_{x_\beta x_\gamma}(t) = 1 \qquad (t > 0).$$

Choose $r \geq 1$ such that $\beta_r \neq \gamma_r$. Consider blocks T_k with $\lambda_k = D_r$. On every such block and every $n \in T_k$, one of the two orbit points lies in $B_{n,1}$ and the other lies in $B_{n,2}$. Since $M_n = 2$ on D_r-blocks,

$$\operatorname{dist}(B_{n,1}, B_{n,2}) > \delta_2(\alpha),$$

where α is the scale of the fixed strict scheme. Put

$$t_0 = \delta_2(\alpha).$$

Then the two points are not t_0-close at any of the L_k iterates belonging to this block. Therefore

$$\frac{1}{S_k + L_k} \#\{0 \leq j < S_k + L_k : |G^j(x_\beta) - G^j(x_\gamma)| < t_0\} \leq \frac{S_k}{S_k + L_k}.$$

Along the infinitely many D_r-blocks, the right-hand side tends to 0. Hence

$$\psi_{x_\beta x_\gamma}(t_0) = 0.$$

The pair (x_β, x_γ) is DC1-scrambled. $\square$

9 Entropy and full metric mean dimension

We first use the E_m-blocks to obtain entropy lower bounds at the precise scales $\delta_m(\alpha)$.

Proposition 9.1 (Entropy lower bounds from E_m-blocks)**.** *Let $g_{1,\infty} \in \mathcal{U}(W_\bullet)$, and let α be the scale of a fixed strict scheme. Then for every $m \geq 2$,*

$$H_g(\delta_m(\alpha)) \geq \log m.$$

Consequently,

$$h(g_{1,\infty}) = \infty.$$

Moreover, the separated orbit segments witnessing these lower bounds may be chosen so that their coded positions at time n lie in W_n.

Proof. Fix $m \geq 2$. Since E_m occurs infinitely often in the label sequence, there are infinitely many blocks T_k with $\lambda_k = E_m$. Fix such a block. Its time interval is

$$T_k = \{S_k + 1, \ldots, S_k + L_k\},$$

and during this block $M_n = m$.

For every word

$$u = (u_{S_k+1}, \ldots, u_{S_k+L_k}) \in \{1, \ldots, m\}^{L_k},$$

choose a finite itinerary up to time $S_k + L_k$ by prescribing the symbols u_n on the block T_k and prescribing symbol 1 outside the block. By Corollary 7.2, there exists a point $x_u \in J_1$ such that

$$G^{n-1}(x_u) \in B_{n,u_n} \subset W_n \qquad (n \in T_k).$$

If $u \neq v$, choose $n \in T_k$ with $u_n \neq v_n$. Then

$$G^{n-1}(x_u) \in B_{n,u_n}, \qquad G^{n-1}(x_v) \in B_{n,v_n}.$$

The two branch intervals are separated by more than $\delta_m(\alpha)$, so

$$|G^{n-1}(x_u) - G^{n-1}(x_v)| > \delta_m(\alpha).$$

Since $n - 1 < S_k + L_k$, the set

$$E_k = \{x_u : u \in \{1, \ldots, m\}^{L_k}\}$$

is $(S_k + L_k, \delta_m(\alpha))$-separated. The points x_u are distinct, because one point cannot realize two finite itineraries which differ at some time. Hence

$$\mathrm{card}(E_k) = m^{L_k},$$

and

$$s_{S_k+L_k}(g_{1,\infty}, \delta_m(\alpha)) \geq m^{L_k}.$$

Taking logarithms and dividing by $S_k + L_k$ gives

$$\frac{1}{S_k + L_k} \log s_{S_k+L_k}(g_{1,\infty}, \delta_m(\alpha)) \geq \frac{L_k}{S_k + L_k} \log m.$$

Along the infinitely many E_m-blocks, $S_k/L_k \to 0$, so

$$H_g(\delta_m(\alpha)) \geq \log m.$$

Since m is arbitrary,

$$h(g_{1,\infty}) = \sup_{\varepsilon>0} H_g(\varepsilon) = \infty.$$

The window assertion follows from $B_{n,u_n} \subset W_n$. □

Lemma 9.2 (Universal interval upper bound)**.** *For every $f_{1,\infty} \in F(I)$,*

$$\overline{\mathrm{mdim}}_M(f_{1,\infty}) \leq 1.$$

Proof. Fix $0 < \varepsilon < 1$ and $N \geq 1$. Consider the orbit map

$$\Gamma_N : I \to I^N, \qquad \Gamma_N(x) = (F^0(x), F^1(x), \ldots, F^{N-1}(x)).$$

If $E \subset I$ is (N, ε)-separated for $f_{1,\infty}$, then $\Gamma_N(E)$ is ε-separated in I^N with respect to the sup norm. Partition I into $K = \lceil 1/\varepsilon \rceil$ intervals of length at most ε. The corresponding product partition covers I^N by K^N boxes of sup-norm diameter at most ε. Each such box contains at most one point of an ε-separated set, because two points in the same box have sup-norm distance at most ε, whereas separated points have distance strictly larger than ε. Therefore

$$s_N(f_{1,\infty}, \varepsilon) \leq \left\lceil \frac{1}{\varepsilon} \right\rceil^N.$$

It follows that

$$H_f(\varepsilon) \leq \log \left\lceil \frac{1}{\varepsilon} \right\rceil.$$

Dividing by $\log(1/\varepsilon)$ and letting $\varepsilon \downarrow 0$ gives

$$\overline{\mathrm{mdim}}_M(f_{1,\infty}) \leq 1.$$

□

Proposition 9.3 (Full metric mean dimension). *Every system $g_{1,\infty} \in \mathcal{U}(W_\bullet)$ satisfies*

$$\underline{\mathrm{mdim}}_M(g_{1,\infty}) = \overline{\mathrm{mdim}}_M(g_{1,\infty}) = 1.$$

Proof. The upper bound follows from Lemma 9.2. We prove the lower bound.

Fix a strict scheme for $g_{1,\infty}$ with scale α. Put

$$c_\alpha = \frac{\alpha}{64K_\alpha}.$$

Then

$$\delta_m(\alpha) = \frac{c_\alpha}{m}.$$

By Proposition 9.1, for every $m \geq 2$,

$$H_g\left(\frac{c_\alpha}{m}\right) \geq \log m.$$

Let $\varepsilon > 0$ be sufficiently small, and choose $m = m(\varepsilon) \geq 2$ such that

$$\frac{c_\alpha}{m+1} \leq \varepsilon \leq \frac{c_\alpha}{m}.$$

Since $H_g(\varepsilon)$ is nonincreasing as a function of ε,

$$H_g(\varepsilon) \geq H_g\left(\frac{c_\alpha}{m}\right) \geq \log m.$$

Also $\varepsilon \geq c_\alpha/(m+1)$, and hence

$$\log(1/\varepsilon) \leq \log\left(\frac{m+1}{c_\alpha}\right).$$

Therefore

$$\frac{H_g(\varepsilon)}{\log(1/\varepsilon)} \geq \frac{\log m}{\log((m+1)/c_\alpha)}.$$

As $\varepsilon \downarrow 0$, one has $m(\varepsilon) \to \infty$, and the right-hand side tends to 1. Hence

$$\underline{\mathrm{mdim}}_M(g_{1,\infty}) \geq 1.$$

Together with Lemma 9.2, this gives both equalities. □

10 Proofs of the main results

Proof of Theorem 2.1. Fix dense open windows $(W_n)_{n\geq 1}$. Let $\mathcal{U}(W_\bullet)$ be the set of systems admitting a strict $W_\bullet$-windowed horseshoe scheme. Lemma 5.2 shows that $\mathcal{U}(W_\bullet)$ is open in $F(I)$. Lemma 6.1 shows that it is dense in $F(I)$.

The same dense implantation lemma also shows that, for every $f_{1,\infty} \in F_s(I)$ and every relative D-neighborhood of $f_{1,\infty}$ in $F_s(I)$, there is a point of $\mathcal{U}(W_\bullet) \cap F_s(I)$ in that neighborhood. Hence $\mathcal{U}(W_\bullet) \cap F_s(I)$ is dense in $F_s(I)$. Its relative openness follows from the openness of $\mathcal{U}(W_\bullet)$ in $F(I)$.

Now take any $g_{1,\infty} \in \mathcal{U}(W_\bullet)$. Proposition 8.1 gives a DC1 scrambled set whose coded orbits visit W_n at time n. Proposition 9.1 gives infinite topological entropy and the windowed separated orbit segments. Proposition 9.3 gives

$$\underline{\mathrm{mdim}}_M(g_{1,\infty}) = \overline{\mathrm{mdim}}_M(g_{1,\infty}) = 1.$$

Thus every asserted conclusion holds. □

Proof of Corollary 2.2. Apply Theorem 2.1 with $W_n = I$ for every n. The desired class contains the dense open set $\mathcal{U}(I_\bullet)$, where $I_\bullet = (I, I, I, \ldots)$. Therefore its complement is contained in the closed nowhere-dense set $F(I) \setminus \mathcal{U}(I_\bullet)$, and so the desired class is residual. The proof in the relative space $F_s(I)$ is identical. □

11 Additional comments

Remark 11.1 (Why the strict version is robust)**.** *The exact equality $g_n(B_{n,i}) = J_{n+1}$ from [10] is enough for density but not for openness, because an arbitrarily small perturbation can destroy exact endpoint matching. The strict inequalities in Definition 5.1 leave a margin. The intermediate value theorem then gives the same covering relation after every perturbation of size less than that margin. This is the whole mechanism behind the passage from density to open density.*

Remark 11.2 (Why this does not contradict non-openness in convergent spaces)**.** *Balibrea and Rucká show that DC1 systems form neither an open nor a closed set in a pointwise-convergent interval space. The present theorem is about the full space $F(I)$, where no convergence condition is imposed on the tail. The constructed strict horseshoe uses this freedom by inserting a controlled local coding relation at every time n.*

Remark 11.3 (Metric mean dimension versus infinite entropy)**.** *Infinite entropy means that $H_g(\varepsilon)$ is unbounded as $\varepsilon \downarrow 0$. Full metric mean dimension is stronger: it says that $H_g(\varepsilon)$ grows at the maximal interval rate, namely like* $\log(1/\varepsilon)$ *up to first order. Proposition 9.3 shows that the moving horseshoes do not merely make the entropy infinite; they make the small-scale orbit complexity maximal.*

References

[1] F. Balibrea and L. Rucká, *Density of distributional chaos in non-autonomous systems*, Topology Appl. **380** (2026), 109735.

[2] F. Balibrea, J. Smítal and M. Štefánková, *The three versions of distributional chaos*, Chaos Solitons Fractals **23** (2005), 1581–1583.

[3] F. Balibrea, J. Smítal and M. Štefánková, *On generic properties of nonautonomous dynamical systems*, Internat. J. Bifur. Chaos Appl. Sci. Engrg. **28** (2018), no. 8, 1850102.

[4] J. Dvořáková, *Chaos in nonautonomous discrete dynamical systems*, Commun. Nonlinear Sci. Numer. Simul. **17** (2012), 4649–4652.

[5] S. Kolyada, M. Misiurewicz and Ł. Snoha, *Topological entropy of nonautonomous piecewise monotone dynamical systems on the interval*, Fund. Math. **160** (1999), 161–181.

[6] S. Kolyada and Ł. Snoha, *Topological entropy of nonautonomous dynamical systems*, Random Comput. Dynam. **4** (1996), 205–233.

[7] E. Lindenstrauss and B. Weiss, *Mean topological dimension*, Israel J. Math. **115** (2000), 1–24.

[8] B. Schweizer and J. Smítal, *Measures of chaos and a spectral decomposition of dynamical systems on the interval*, Trans. Amer. Math. Soc. **344** (1994), 737–754.

[9] H. Shao, Y. Shi and H. Zhu, *On distributional chaos in non-autonomous discrete systems*, Chaos Solitons Fractals **107** (2018), 234–243.

[10] Y. Shmalo, *Dense simultaneous distributional chaos and infinite entropy in the full space of non-autonomous interval systems*, preprint, 2026.

[11] M. Štefánková, *Inheriting of chaos in uniformly convergent nonautonomous dynamical systems on the interval*, Discrete Contin. Dyn. Syst. **36** (2016), 3435–3443.